

DEMANDSHOCK

Crisis-Aware Demand Forecasting & Inventory Intelligence Platform

Technical Project Report

Design, implementation, measured results and independent verification

Attribute	Value
Platform version	1.0.0
Run mode	Full dataset (all 30,490 item-store series)
Primary model	Global LightGBM (Tweedie), one model per horizon bucket
Selected feature set	B (+ Calendar & Price)
Data coverage	2011-01-29 to 2016-06-19
Model trained	2026-08-05T00:19:03+00:00
Configuration hash	f87039f3db9b
Automated tests	135 passing, 0 skipped
Verification status	Independently audited (read-only)

Every figure in this report is read directly from the pipeline artifacts produced by the full-scale run. No number has been typed by hand, estimated, or illustrated.

1. Executive Summary

DemandShock is a decision-support platform for retail demand planning. It forecasts daily demand for every product in every store, detects where actual demand has broken away from what the model expected, attaches the real-world conditions that coincided with those breaks, and translates the forecasts into transparent inventory planning arithmetic.

The distinguishing idea is that the forecast residual is treated as a signal in its own right, not merely as an error term. Conventional tools report an average accuracy figure; they do not tell a planner which specific series broke, how badly, what else was happening at the time, or what it is worth in dollars.

1.1 Headline measured results

Result	Value	Basis
Series modelled	30,490	M5 item × store
Item-days processed	46,881,677	after release filtering
Forecast accuracy (WAPE)	0.7198	28-day held-out window
Scaled error (RMSSE)	0.7550	below 1.0 beats naive
Improvement vs seasonal naive	19.1%	same information set
Near-term accuracy (steps 1–7)	0.7226	was 0.7519
Weekly-grain WAPE	0.3683	same forecast, weekly totals
Total-demand WAPE	0.0684	all series summed per day
Interval coverage	79.4%	target 80%, n=853,624
Shock episodes detected	52,922	out-of-sample window
Estimated revenue exposure	\$3,046,781	real prices, 28 days
Data quality checks	63 / 63 passed	0 warnings, 0 failures

1.2 The most important finding

The project set out to test whether external crisis and economic context improves demand forecasting. Measured honestly across identical rolling-origin folds, the answer is no. Adding FEMA disaster context changed WAPE by **0.0002**, smaller than the **0.0088** fold-to-fold variation. Adding FRED economic context made accuracy slightly worse. The selection rule therefore chose the simpler feature set, and both sources were retained for explaining shocks rather than for point-forecast accuracy.

This is reported prominently inside the application itself, auto-generated from the numbers rather than written by hand. A platform that quietly buried this result would be less trustworthy, not more.

What this platform is, and is not

It is decision support: forecasts, deviation detection, context and planning arithmetic for a human planner.
It is not an autonomous replenishment system, it does not claim causation between disasters and demand, and it never presents an assumption as an observation.

2. Data Sources and Integrity

The platform uses only real, publicly documented datasets. No synthetic records, simulated crises, fabricated inventory positions or placeholder metrics exist anywhere in the pipeline, the application, or this report.

Source	Contribution	Grain
M5 Forecasting – Accuracy	Daily unit sales, weekly sell prices, retail calendar with events and SNAP flags	Item × store × day
FEMA OpenFEMA v2	Disaster declarations: type, dates, designated counties	Disaster × county
FRED / BLS	Monthly state unemployment for California, Texas, Wisconsin	State × month

2.1 Volume actually processed

Quantity	Value
Item-store series	30,490
Distinct items / stores	3,049 / 10
Categories / departments / states	3 / 7 / 3
Item-days after release filtering	46,881,677
Weekly price records	6,841,121
Calendar days covered	1,969 (2011-01-29 to 2016-06-19)
Zero-sale share of item-days	59.6%
FEMA disasters in window (CA/TX/WI)	116
FRED monthly observations	1,818

Demand is highly intermittent: **59.6%** of item-days record zero sales. This single fact drives most of the modelling decisions that follow, from the choice of objective function to the way metrics are interpreted and the guards built into shock detection.

The ingestion stage runs 63 automated checks and refuses to proceed if any fails. On the reported run: 63 passed, 0 warnings, 0 failures.

2.2 Deliberate exclusions

- NOAA GHCN-Daily weather extracts are present on disk but never read. Weather is out of scope in this version; no feature, page or endpoint depends on it.
- Rows before a series' first priced week are dropped rather than treated as zero demand — the product did not exist in that store yet. The count removed is reported, never applied silently.

3. Forecasting Methodology

3.1 The governing invariant

Invariant

Every feature for target date t is computable from information available at the forecast origin O , or is genuinely known in advance (retail calendar, SNAP schedule, published weekly price).

Demand lags and rolling windows are shifted by the bucket's own shift, and FEMA and FRED tables are joined at the origin — the state of the world as it was known when the forecast was made, not as it later turned out.

3.2 Horizon buckets

A 28-day path is issued once at origin O , so a model serving steps $1..S$ may read data only up to O — a shift of S relative to the target date. The platform originally froze every step at 28 days so a single model could serve all horizons. That had a perverse consequence: the 1-day-ahead forecast read demand from 27 days earlier, while the 28-day-ahead forecast read the freshest day available. Accuracy was consequently flat across the horizon — the opposite of what a forecaster should do.

Bucket	Steps	Origin shift	Newest input at its last step
H1	1–7	7 days	the forecast origin
H2	8–14	14 days	the forecast origin
H3	15–28	28 days	the forecast origin

The safety rule is **shift** $\geq$ **max_step**: at a bucket's last step the newest input is exactly the origin. Three automated tests enforce it — one checks every bucket satisfies the rule and that the buckets tile steps 1 to 28 exactly once, and a parameterised test re-proves the perturbation invariant at each shift, each with a control showing the test cannot pass vacuously.

Measured effect

Forecast steps	Single 28-day freeze	Horizon buckets	Improvement
1–7 days ahead	0.7519	0.7226	3.9%
8–14 days ahead	0.7385	0.7233	2.1%
15–28 days ahead	0.7274	0.7274	unchanged
All 28 days (WAPE)	0.7305	0.7198	1.5%
All 28 days (RMSSE)	0.7606	0.7550	0.7%

Two controls confirm the result is real rather than an artefact. Steps 15–28 are unchanged, exactly as predicted, because their shift did not move — 28 was already the correct value for that bucket. And the naive benchmarks are identical to four decimal places, as they must be since they use no features at all.

The accuracy curve now slopes upward with horizon: near-term forecasts are the sharpest.

What the experiment showed before the change was adopted

An isolated experiment on one fold compared shifts directly. Fresher rolling windows produced the entire gain; fresher individual day-lags added nothing measurable. On intermittent demand a single day's lag is noise where a rolling mean is signal. The change is therefore a per-bucket shift, not a feature redesign — a smaller and far safer edit.

3.3 Why direct rather than recursive

With over half of item-days at zero, feeding fractional predictions back into integer-heavy lag features compounds distribution shift across 28 steps and would contaminate the very residuals the shock module depends on. Direct forecasting also means training and inference share one feature function, so there is no second implementation to drift out of sync.

3.4 Validation design

Validation is chronological throughout. No random splitting exists anywhere in the codebase.

Split	Dates	Role
Fold F1	2016-02-01 to 2016-02-28	Ablation, residual pool
Fold F2	2016-02-29 to 2016-03-27	Ablation, residual pool
Fold F3	2016-03-28 to 2016-04-24	Ablation, residual pool
Holdout	2016-04-25 to 2016-05-22	Headline results, opened once
Forward	2016-05-23 to 2016-06-19	Forward forecast, no actuals

A metric deliberately not claimed

The official M5 WRMSSE aggregate is not implemented, so the name is never used. An automated test scans the entire codebase to ensure no component claims it.

4. Measured Results

All figures come from the held-out window 2016-04-25 to 2016-05-22, excluded from training entirely and opened once after the feature set had been chosen.

4.1 Model comparison

Model	WAPE	RMSSE	MAE	RMSE	Bias	sMAPE
LightGBM (set B)	0.7198	0.7550	1.0385	2.1493	-0.0655	139.8
Seasonal naive (lag 7)	0.8622	0.9976	1.2440	2.6769	-0.0736	83.1
Seasonal naive (lag 28)	0.8900	1.0338	1.2840	2.8292	-0.0391	84.0
Naive (last observed day)	0.9516	1.0017	1.3730	2.8936	0.1319	85.7

LightGBM reduces WAPE by **19.1%** against seasonal-naive at lag 28, the benchmark that forecasts from the same 28-day-old information. Its RMSSE of **0.7550** is below 1.0, meaning it also beats a one-day naive forecast measured against each series' own training history.

The sMAPE column illustrates why metric choice matters on intermittent demand. The naive models score better on sMAPE precisely because they predict exact zeros, which are scored as perfect on the majority of days. WAPE and RMSSE are the trustworthy comparisons, and the application labels sMAPE accordingly.

4.2 Feature attribution

Attribution uses exact TreeSHAP values computed natively by LightGBM; the optional shap package is not required.

Feature family	Share of model gain
Demand history	87.79%
Product / store identity	9.17%
Calendar and events	1.87%
Price	1.17%

Recent demand history dominates, the expected outcome for short-horizon retail forecasting. Attribution measures reliance, not direction: the values are unsigned, and the application states this rather than asserting a direction the artifacts cannot support.

5. Ablation: Do External Signals Help?

Four cumulative feature sets were evaluated across identical rolling-origin folds with the same seed, parameters, training rows and eligible series. The only difference between arms is which features are present.

Arm	Feature set	Features	WAPE	SD across folds
A	Demand history and identity only	19	0.7591	0.0076
B	Plus calendar and price	36	0.7472	0.0088
C	Plus FEMA disaster context	43	0.7470	0.0096
D	Plus FRED economic context	47	0.7473	0.0093

5.1 Verdict

- **Calendar and price features improved accuracy.** WAPE fell by 0.0119 (1.57% relative), consistently across folds. This is the one external addition that clearly earns its place.
- **FEMA disaster context did not materially help.** WAPE moved by 0.0002, far smaller than the 0.0088 fold-to-fold spread. A difference this small cannot be distinguished from noise.
- **FRED economic context did not help.** WAPE rose by 0.0003. Monthly state unemployment is too slow-moving to explain daily deviations in a 28-day horizon.
- **Selection outcome.** The mechanical rule chose the smallest feature set within 0.3% relative of the best mean WAPE, which is arm B. The simpler model was selected on the evidence, not on preference.

The ablation was re-run under horizon bucketing rather than assuming the earlier verdict still held. Every arm improved by a similar margin and the conclusion was unchanged.

Why the negative result is valuable

A common failure mode in applied data science is to add an interesting external dataset and then find a way to justify keeping it. This study measured the contribution under a fair protocol, found it immaterial, reported it in the application and in the generated results file, and let the selection rule act on it. The external sources are retained where they genuinely add value — as context for interpreting shocks — and not where they do not.

6. Demand Shock Intelligence

A shock is not simply high demand. It is a day on which actual demand departed from what the model could reasonably have expected, in a way that is large relative to that series' own recent forecast error, and/or persistent, and/or accompanied by a change in volatility or demand level.

Shock residuals are taken from arm B, which contains no FEMA or FRED features. This is deliberate: if crisis features were in the model, crisis-driven deviations would be partly absorbed into the prediction and would disappear from the residual, blinding the detector to exactly what it exists to find.

6.1 Score construction

Signal	Weight	What it measures
Standardised residual	35	Deviation relative to the series' own recent error
Percentage deviation	20	Deviation relative to recent average demand
Persistence	20	Consecutive days deviating in the same direction
Volatility ratio	10	Short-term error spread against the longer-term spread
Level shift	15	Change in the demand level itself, not just the error

Piecewise-linear squashing was chosen over smooth functions so every point of the score is attributable to a component and the arithmetic reconciles by hand in the explanation panel. Guards against false positives include a low-volume damping multiplier, a hard zero for a zero-sale day against a near-zero forecast, an eligibility gate, a warm-up cap, and residual dispersion frozen at its pre-episode value so a long episode cannot inflate its own baseline.

6.2 Measured detection results

Scored over 21,900 eligible series and 2,452,463 scored series-days, 52,922 episodes were filed with a median duration of 4 days.

Classification	Episodes	Share
Volatility Shock	23,175	43.8%
Demand Surge	15,454	29.2%
Demand Collapse	12,153	23.0%
Persistent Over-forecast	1,548	2.9%
Regime Shift	417	0.8%
Possible Regime Shift (window truncated)	127	0.2%
Persistent Under-forecast	48	0.1%

13.6% of episodes coincided with a FEMA disaster declaration active in the same state. The platform reports this as a coincidence rate and never as a causal effect.

The wording rule, enforced by test

Permitted: a FEMA disaster declaration was active within the state during this period. Not permitted: the disaster caused demand to increase. FEMA declarations are county-scoped while M5 discloses only a store's state, so the relationship is a state-level temporal coincidence. An automated test scans every generated context sentence for causal language.

7. Reading a Forecast Correctly

A recurring source of misreading is the daily chart of a slow-moving product. The forecast appears as a smooth line near half a unit while actual demand jumps between zero and three. The forecast looks wrong. It is not — and understanding why is essential to using the platform, so the interface now explains it at the point of confusion.

7.1 Why a point forecast cannot track a spiky series

A product selling a few units a week contains no information about which day the sale will land. The best possible point forecast is its expected value — roughly the weekly rate divided by seven — and that line will never sit on the spikes. Demanding otherwise is asking the model to predict which customer walks in on which day. Across the held-out window 59.6% of item-days record zero sales, so this is the normal case rather than an edge case.

7.2 The same forecast, judged at three grains

View	WAPE	What it answers
Daily, item level	0.7198	Will this product sell on this exact day?
Weekly, item level	0.3683	Is the weekly requirement right? (ordering grain)
Total demand per day	0.0684	Is overall demand right? (capacity, staffing)

Nothing changes between these rows except the aggregation. The identical forecast is roughly **twice as accurate** at the weekly grain a planner actually orders on, and tracks total demand to within **6.8%**. The daily item-level figure is dominated by the irreducible randomness of which day a purchase falls on, not by model error.

7.3 Uncertainty is the answer, not the point forecast

For an intermittent series the honest output is a range, not a number. The platform publishes empirical P10–P90 intervals derived from out-of-sample residuals. A slow mover with an expected 0.5 units a day typically carries an interval spanning roughly zero to two units — precisely the statement a planner needs.

Interval property	Measured on the holdout
Design target	80%
Empirical coverage	79.4%
Actuals below P10	9.7%
Actuals above P90	10.8%
Evaluation points	853,624

A defect this analysis uncovered

Reviewing these charts revealed that the P10–P90 band had never rendered in the application. PyArrow infers a partitioned dataset's schema from its first fragment; the backtest folds carry no interval columns while the holdout does, so reading the forecast store silently dropped those columns everywhere. No error was raised — the chart simply found no interval present and drew nothing. The loader now unifies schemas across fragments, and a regression test pins the behaviour.

7.4 Accuracy is not uniform across the assortment

Sales velocity	WAPE	Mean units/day	Share of units
Top seller	0.5110	6.32	43.8%
Fast mover	0.7070	1.89	26.3%
Steady	0.9300	0.89	18.6%
Slow mover	1.1580	0.49	8.5%
Very slow	1.3650	0.28	2.9%

A top seller is forecast more than twice as accurately as a very slow mover. Any assessment of the platform should therefore state which slice it refers to; a single headline figure averages two very different regimes.

8. Product Identification

M5 anonymises product identities. The source data contains no product names — only codes such as FOODS_3_090 — because the retailer removed them before release. Inventing names would have made the platform's central claim of using only real data false, and would be the first thing an examiner questions.

Each series is therefore described by attributes measured from its own history:

Example descriptor

Foods - Dept 3 | Value | Top seller in Foods | 130.8/day (\$1.38)

Component	Derivation
Department	The real M5 hierarchy
Price band	Median sell price, ranked within its category
Velocity	Total units sold, ranked within its category
Units per day	Measured mean daily sales
Price	Real median sell price

Ranks are taken within a category and across all stores, so the same product can be described differently in different stores. That is correct: each item-store pair is its own demand series.

A correction worth recording

Because velocity is ranked within a category, an early version labelled a hobbies product selling 0.78 units a day a “Fast mover”, while the same words in groceries meant 2.23 units a day. Readers reasonably interpreted the label as volume. The descriptor now names the category it is relative to and states the absolute rate, so “Fast mover in Hobbies · 0.7/day” cannot be confused with “Top seller in Foods · 130.8/day”.

These are display labels only. One test asserts that none of the descriptor fields appears in the model's feature list, so they cannot influence a forecast. A second scans the vocabulary for words that could not be derived from M5, so a future edit cannot quietly introduce fabricated names.

9. Inventory and Business Impact

The constraint that shapes this module

M5 contains no inventory records: no on-hand stock, no lead times, no service levels and no costs. Rather than inventing values, the module separates what can be measured from what must be supplied by the user, and labels the difference in the interface.

9.1 Measured from real data

Quantity	Value
Series covered	30,490 (30,463 eligible for planning)
Actual units sold in the holdout window	1,231,648
Forecast units over the same window	1,150,979
Revenue represented at real sell prices	\$3,900,555
Under-forecast exposure	\$1,656,859
Over-forecast exposure	\$1,389,922
Total estimated revenue exposure	\$3,046,781

Estimated Revenue Exposure dollarises forecast error using real M5 sell prices. It is deliberately not called lost revenue: M5 records units sold, so demand that was never satisfied is unobservable in the source data.

9.2 Requires user-supplied operating assumptions

Quantity	Formula
Lead-time demand	Sum of the daily forecasts over the lead time
Safety stock	$z \times \sigma_{\text{daily}} \times \sqrt{\text{lead_time}}$
Reorder point	Lead-time demand + safety stock
Days of cover	On-hand units / mean daily forecast
Projected stockout	First day cumulative forecast exceeds on-hand units

Sigma is the standard deviation of that series' own out-of-sample daily forecast errors, measured across the backtest folds rather than assumed. Safety stock, reorder point, days of cover and projected stockout appear only after the user supplies a lead time and explicitly selects a service level.

9.3 Separating technical results from business assumptions

The measured technical results are the accuracy improvement over benchmarks, the interval coverage, the ablation finding, the shock statistics and the dollar-valued exposure at real prices. Any translation of

those into inventory savings, stockout reduction or return on investment is an assumption layer requiring service levels, lead times and holding costs the data does not contain. The platform does not make that translation, and neither does this report.

10. Engineering Quality and Verification

The suite comprises 135 automated tests, all passing with none skipped in the reference environment.

Area	What is proven
Temporal leakage	Perturbing actuals inside a bucket's origin window leaves every feature bit-identical, with a control test proving the check is not vacuous
Horizon buckets	Every bucket satisfies $\text{shift} \geq \text{max_step}$ and the buckets tile steps 1–28 exactly once
External signal timing	FEMA flags stay off before a declaration was issued; FRED respects the publication boundary to the day
Data provenance	The melt, price ingest and FEMA deduplication are checked against the raw source CSV files
Metrics	Hand-computed values, degenerate cases, pooled versus averaged aggregation
Shock scoring	Score reconciliation, determinism, guards, sigma freezing, and all seven classifications driven end-to-end
Descriptors	Derived from measured data, never model features, no invented names
Prediction intervals	The band survives the partitioned read (regression test)
Inventory	Formula correctness, lead-time clamping, nothing produced without input
Service layer	Response schemas and the 200, 404, 422 and 503 paths
Graceful degradation	Every page renders guidance rather than a traceback when artifacts are absent

10.1 Independent verification

The completed project was subjected to a read-only audit that recomputed the headline metrics from the raw prediction artifacts without using project code, reconstructed shock scores from their stored components, traced processed tables back to the original CSV files, and reviewed the code for leakage and the interface for misleading presentation.

Verification	Outcome
Holdout metrics recomputed independently	Matched to every printed digit
Interval coverage recomputed	Matched; no ordering violations
Shock score arithmetic reconstructed	Reconciled on every scored row
Melt and price ingest versus raw CSV	Exact match on all sampled cells
Ablation fairness	Identical training rows and series in every arm
Causal language scan	No violations
Pipeline determinism	Independent re-runs reproduced results to six decimals

11. Limitations and Responsible Use

These limitations are documented in the repository, displayed inside the application, and listed by the service layer.

11.1 Data limitations

- M5 contains no inventory records. Safety stock, reorder points and days of cover depend on operating assumptions the user supplies.
 - M5 records units sold, not demand. Unmet demand is unobservable, which is why the platform reports exposure rather than lost revenue.
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- M5 anonymises products; descriptors are measured attributes, not names.
 - FEMA declarations are county-scoped while M5 discloses only a store's state.
 - FRED unemployment is monthly and cannot explain a daily deviation.
 - Weather data is out of scope in this version.

11.2 Methodological limitations

- Each horizon bucket sees data only as of its own origin, so a 7-day-ahead forecast does not use the most recent six days — the deliberate cost of issuing one 28-day path from a single origin.
- FEMA active flags switch off using recorded incident end dates, knowable only retrospectively; the onset is strictly point-in-time.
- The model and the naive benchmarks are scored on slightly different rows, because the model requires a feature warm-up period the benchmarks do not.
- Shock detection is retrospective: residuals exist only where the model forecast out-of-sample, so every shock shown is a measured historical deviation.
- Hierarchical aggregation is bottom-up; no formal forecast reconciliation.
- No causal inference is attempted. External signals are association only.
- The demand shock score is a project-defined metric, not an industry standard.

11.3 Responsible use

The platform is decision support. Forecast uncertainty is displayed rather than hidden, poorly performing segments are surfaced rather than buried, and the ablation verdict is generated from the numbers so it cannot be quietly softened. Human review belongs between any output here and an ordering decision.

12. Conclusion

DemandShock demonstrates an end-to-end applied data science workflow: data engineering at scale, time-series forecasting under strict temporal discipline, anomaly and regime detection, external data integration evaluated on its merits, explainability, inventory decision support, software engineering

practice, and executive communication.

The forecasting model measurably outperforms the benchmark that shares its information set, its prediction intervals are well calibrated at scale, and its shock detection produces auditable, individually explainable episodes. Equally important, the project reports honestly where its hypothesis failed: external crisis and economic context did not materially improve point forecasts, the selection rule acted on that evidence, and the finding is displayed rather than concealed.

The result is a platform whose numbers can be traced from the interface back through the artifacts to the original source files, and which states its limitations as clearly as its results.

End of report.